Stochastic Calculus and Option Pricing
Lecture 9 Exercises

9 Week 9 — Vanilla Options in Practice

Focus. A short theoretical variation on contract units that remains valid without choosing market numbers.

Exercise 9.1 (A stock split changes units, not economics)**.** Suppose one old share is replaced by $a > 0$ new shares, so that the quoted stock price and strike become $S' = S/a$ and $K' = K/a$.

(a) Compare the old call payoff $(S_T - K)^+$ with the payoff on a new calls struck at K/a. Explain why the option position must also be rescaled by a.

(b) If $C(S, K)$ is the old call value and $C'(S', K')$ the value per new call, derive the relation between them by no-arbitrage.

(c) Use the chain rule to relate the old and new Delta and Gamma. Which sensitivity is invariant to the change of share denomination?